

Flood Catastrophe Reinsurance & Capital Optimisation

Gulf Coast reference portfolio (FL, TX, LA, MS, AL), 2025 exposure: what to buy, what it costs, what it removes

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Generated by `make report from outputs/ (git 0182dd7-dirty)`. All money in USD, 2025 exposure basis. Prices are indicative technical premiums, not market quotes. Economic capital (EC) is a 99.5% one-year proxy for this portfolio's insurance risk, not a Solvency II SCR.

1. Executive summary

What to buy. 20% quota share + 1-in-5 to 1-in-200 Cat XoL, 80% placed, no reinstatement. In the base model it reduces the 99.5% one-year economic capital proxy from USD 25,325m to USD 4,631m, a capital relief of **81.7%**, for an expected net cost of **USD 1,215m a year** (premium USD 1,966m less expected recoveries of USD 752m). That is an implied cost of capital of **5.9%**, below the 10% hurdle assumed for the insurer's own capital.

Why. The programme sits on the cost-capital frontier of 255 candidate programmes and is the Balanced choice at a 10% hurdle: every additional step of protection up to it releases capital at a marginal cost below the hurdle, and the next step (20% quota share + 1-in-5 to 1-in-200 Cat XoL, 100% placed, no reinstatement) costs 10.1% per unit of capital released. Equivalently, it minimises the total cost of risk (net reinsurance cost + 10% × net EC): USD 1,678m against USD 2,532m with no reinsurance and USD 2,256m for the cheapest programme that releases at least 25% of capital.

How much to trust it. The exact contract held in 4 of 9 stress scenarios and its layer structure in 9 of 9; the others change only the placement share. Two things drive the result and deserve management's attention. First, the reinsurer's diversification: with none ($d = 1$), reinsurance capital costs more than equity and the same rule picks no reinsurance. Second, the tail rests on few events: without Hurricane Harvey gross EC falls by 57%, and the year bootstrap gives a 90% interval of 53%–82% for capital relief.

Affordability. The recommended programme's premium is 76% of the portfolio's own premium (USD 2,593m). This is large because, at a 10% hurdle, holding the gross capital would cost USD 2,532m a year, more than the portfolio's expected margin before expenses (USD 1,044m). If that premium is not affordable, the Budget-first programme (1-in-50 to 1-in-100 Cat XoL, 80% placed, no reinstatement) releases 26% of capital for USD 370m a year.

Choice	Programme	Net cost (USD m)	Net EC (USD m)	Relief	Implied CoC
Budget-first	1-in-50 to 1-in-100 Cat XoL, 80% placed, no reinstatement	370	18,852	25.6%	5.7%
Balanced (h = 10%)	20% quota share + 1-in-5 to 1-in-200 Cat XoL, 80% placed, no reinstatement	1,215	4,631	81.7%	5.9%
Protection-first	20% quota share + 1-in-5 to 1-in-200 Cat XoL, 100% placed, no reinstatement	1,447	2,338	90.8%	6.3%

2. Portfolio and catastrophe model

Portfolio. NFIP policies in force in the five Gulf states, written in 2025: 2,892,970 insured units, USD 805bn total insured value (59% in Florida) and USD 2,593m premium. It is a stylised reference portfolio built from NFIP data, not a model of the NFIP's finances. National totals from the same extract (4,677,822 insured units, USD 1.29tn) match the order of magnitude published by the Congressional Research Service.

Events. Claims (499,686 in 2009–2025) are grouped by FEMA's `floodEvent` label and each event is restated to 2025 exposure by the ratio of state written insured value in 2025 to the event year (no CPI, which would double count value growth). Events of at least USD 100m as-if form the catalogue: 20 events in 2009–2023 totalling USD 22.7bn; everything else is attritional (mean USD 142m a year). In the active years one event supplied most of the year's loss, the signature of accumulation risk.

Event	Year	Nominal (USD m)	As-if 2025 (USD m)
Hurricane Harvey	2017	9,056	8,880
Hurricane Helene (holdout)	2024	6,487	6,524
Hurricane Ian	2022	4,867	5,262
Mid-summer severe storms	2016	2,534	2,226
Hurricane Milton (holdout)	2024	1,211	1,218
Hurricane Irma	2017	986	1,039

Loss model. Annual loss = attritional + sum of event losses. Frequency is Poisson with 1.33 events a year (dispersion index 1.14; likelihood-ratio statistic 0.02, far below the 2.71 needed for a negative binomial). Severity is a log-logistic fitted by maximum likelihood with left truncation at USD 100m, chosen by AIC among lognormal, log-logistic and Burr XII after checking that each of the five largest events is unremarkable under the fit. The three fits are within 0.7 AIC points of each other; with only 20 events, generalised Pareto tail estimates were unstable, so GPD is used only as a stress. Each simulated event is capped at $3 \times$ the largest calibration event (USD 26,639m). Attritional loss is bootstrapped from the calibration years.

Simulation. 100,000 years (seed 20261001); VaR 99.5 moves by 0.17% at 200,000 years. Gross mean loss USD 1,549m, 1-in-200 annual loss USD 26,874m, EC USD 25,325m. The 1-in-200 event sits at the event cap, so the far tail is an assumption, and is tested below. The tail beyond the largest observed event therefore comes from the fitted shape and the cap, not from data.

Return period	5	10	20	50	100	200	250
OEP (USD bn)	1.2	2.5	4.6	10.0	18.2	26.6	26.6
AEP (USD bn)	1.8	3.3	5.7	11.4	19.7	26.9	27.1

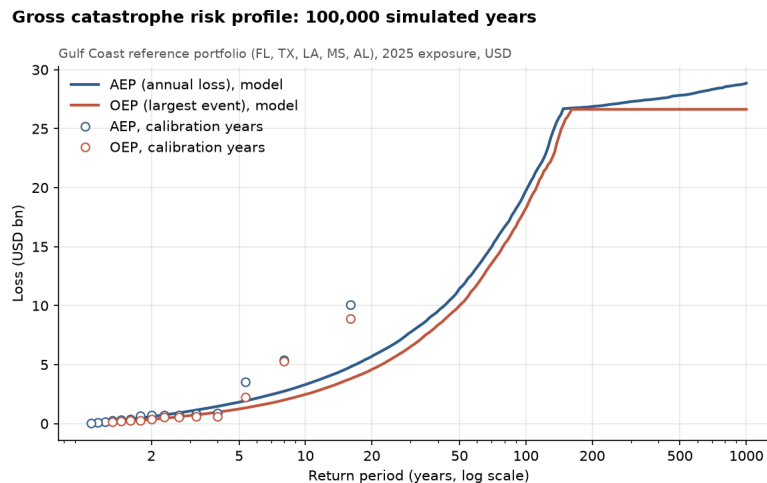


Figure 1: Gross OEP and AEP with the calibration years

3. Reinsurance structure and pricing

Contracts. Quota share (10% or 20%) inures to catastrophe excess of loss, which responds per event to the retained event loss. Layers are designed in return-period terms: attachment at the 1-in-5, 10, 20 or 50 event loss, exhaustion at 1-in-50, 100, 200 or 250, as single layers or two-layer towers split at 1-in-50 or 1-in-100, placed 80% or 100%, with zero or one reinstatement at 100%. The grid has 387 programme IDs and 255 distinct dollar contracts (exhaustion at 1-in-200 and 1-in-250 coincide because both event losses sit at the cap).

Pricing. Each layer's indicative technical premium covers its modelled expected loss (EL), a 10% cost of capital on $0.5 \times$ the reinsurer's standalone capital (TVaR 99.5 minus EL), and a 10% expense load; expected reinstatement premium is part of the reinsurer's income, so the upfront premium is reduced accordingly. The quota share is priced on the same principles with a 3% expense load. The factor d stands for the reinsurer's diversification: in a global book the layer's marginal capital is far below its standalone capital, and without that no reinsurance could ever be cheaper than the insurer's own capital. Historical burning cost is shown beside modelled EL: the 1-in-5 to 1-in-50 layer was hit in 3 of the 15 calibration years and burned USD 843m a year against a modelled EL of USD 510m, a reminder that the fitted tail is not conservative.

Recommended programme.

Treaty	Attach (USD m)	Exhaust (USD m)	Return periods	AP	Share	Re- inst.	Premium (USD m)	ROL	Multi- ple
Quota share 20%	–	–	all losses	–	20%	–	597	–	–
Cat XoL 1	990	21,310	1-in-5 to 1-in-200	20.0%	80%	0	1,369	8.4%	3.1×

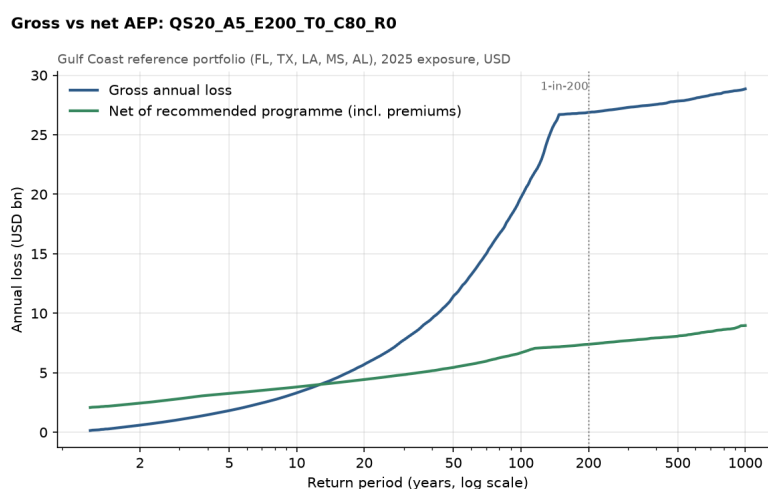


Figure 2: Gross vs net AEP for the recommended programme

4. Capital and programme decision

For every programme, net annual loss = gross loss – recoveries + premiums + reinstatement premiums. A fixed premium raises both VaR and the mean, so it does not change EC; capital relief comes from recoveries in the tail. Net cost = premiums + expected reinstatement premiums – expected recoveries, the price of transferring the risk; implied cost of capital = net cost ÷ capital relief.

Across all programmes the implied cost of capital lies between 5.7% and 8.5%, all below the 10% hurdle, because the reinsurer charges about 5% on capital plus expenses. 79 programmes are efficient (no other programme is both cheaper and needs less capital). Walking up the lower convex hull of the frontier from no reinsurance, each step's marginal cost of capital is:

Hull point	Net cost (USD m)	Net EC (USD bn)	Relief	Marginal CoC
no reinsurance	0	25.3	0%	–
20% quota share only	287	20.3	20%	5.7%
20% quota share + 1-in-50 to 1-in-100 Cat XoL, 80% placed, no reinstatement	583	15.1	40%	5.7%
20% quota share + 1-in-50 to 1-in-100 Cat XoL, 100% placed, no reinstatement	657	13.8	46%	5.8%
20% quota share + 1-in-20 to 1-in-100 Cat XoL, 100% placed, no reinstatement	905	9.6	62%	5.9%
20% quota share + 1-in-5 to 1-in-200 Cat XoL, 80% placed, no reinstatement	1,215	4.6	82%	6.2%
20% quota share + 1-in-5 to 1-in-200 Cat XoL, 100% placed, no reinstatement	1,447	2.3	91%	10.1%

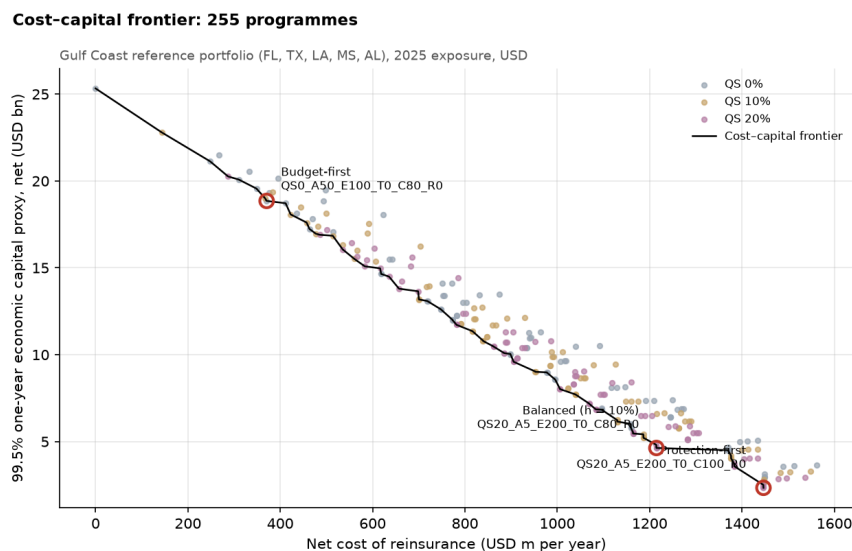


Figure 3: Cost-capital frontier

The Balanced rule stops before the first step whose marginal cost exceeds the hurdle. At 8% it picks `QS20_A5_E200_T0_C80_R0`; at 12%, `QS20_A5_E200_T0_C100_R0`. There is no single optimum: the choice depends on the budget, the risk appetite and the cost of the insurer's own capital, which is why the frontier and the three choices are shown together.

5. Validation, sensitivities and market benchmark

Convergence. The recommended programme's capital relief moves by 0.14% at 200,000 years.

Temporal holdout. Refitted on 2009–2018, the model expected 6.0 events in 2019–2023; 8 occurred (percentile 79%). Hurricane Ian was a 1-in-23 event under that model. The five-year window can reveal a model that is badly off, not confirm a 1-in-200 tail.

Largest event removed. Without Hurricane Harvey, gross EC falls to USD 10,783m (–57%) and the Balanced choice becomes 20% quota share + 1-in-5 to 1-in-50 Cat XoL, 100% placed, no reinstatement; the recommended contract still releases 74% of the smaller capital at 8.1%.

Parameter uncertainty. Resampling the 15 calibration years (200 replicates) gives 90% intervals of USD 4,062m–29,675m for gross EC, 53%–82% for the recommended programme's relief and 5.9%–11.0% for its implied cost of capital.

Stresses and sensitivities (the contract grid is fixed in dollars; each scenario re-prices it and re-applies the Balanced rule):

Scenario	Balanced choice	Same contract	Recommended: relief / CoC
Event frequency $\times 1.25$	<code>QS20_A5_E200_T0_C80_R0</code>	yes	81% / 6.0%
Event severity $\times 1.15$	<code>QS20_A5_E200_T0_C100_R0</code>	placement only	81% / 6.0%
Heavier tail (GPD above median, $\xi + 0.1$)	<code>QS20_A5_E200_T0_C100_R0</code>	placement only	79% / 6.2%
Reinsurer CoC 8%	<code>QS20_A5_E200_T0_C100_R0</code>	placement only	82% / 4.8%
Reinsurer CoC 12%	<code>QS20_A5_E200_T0_C80_R0</code>	yes	82% / 7.0%
Diversification $d = 0.3$	<code>QS20_A5_E200_T0_C100_R0</code>	placement only	82% / 3.6%
Diversification $d = 0.7$	<code>QS20_A5_E200_T0_C80_R0</code>	yes	82% / 8.1%
Cedent hurdle 8%	<code>QS20_A5_E200_T0_C80_R0</code>	yes	82% / 5.9%
Cedent hurdle 12%	<code>QS20_A5_E200_T0_C100_R0</code>	placement only	82% / 5.9%
Event threshold USD 250m	<code>QS20_A5_E200_T0_C100_R0</code>	placement only	81% / 6.0%

Scenario	Balanced choice	Same contract	Recommended: relief / CoC
CPI-only as-if	QS20_A5_E200_T0_C100_R0	placement only	66% / 5.9%
Calibration from 2010	QS20_A5_E200_T0_C80_R0	yes	82% / 5.9%
Lognormal attritional	QS20_A5_E200_T0_C80_R0	yes	82% / 5.9%
Lognormal severity	QS20_A5_E100_T0_C100_R0	no	79% / 7.0%
Event cap 5× largest event	QS0_A5_E200_T0_C100_R0	no	70% / 6.2%
No diversification (d = 1)	QS0_XS0	no	82% / 11.5%

Market benchmark. Compared by attachment probability (4%–9%), technical rates on line of 7.2%–9.0% are below FEMA’s traditional NFIP placements (18.0%, 19.5% and 18.5% in 2023–2025), largely because the model’s layers are wide (low expected loss per unit of limit), while FloodSmart Re’s tranches are thin. Price multiples of expected loss (2.8–4.6×) are comparable with FloodSmart Re cat bond spreads over expected loss (2.7–3.0×). FEMA buys far less protection, attaching at USD 7bn nationally with 9%–26% shares, under a public budget rather than a cost-of-capital objective. The benchmarks cover the national NFIP (all floods for FEMA’s tower, named storms only for FloodSmart Re) and support only this order-of-magnitude comparison; the model was not adjusted towards them.

6. Limitations

- Historical-event model, not a vendor catastrophe model: no hazard, hydraulics or vulnerability modules.
- 15 calibration years: the 1-in-200 tail rests on the severity family and the event cap, with wide parameter uncertainty. Pre-2009 events (Katrina, Ike) are excluded for lack of consistent exposure data.
- As-if restatement adjusts only for state-level insured value: no within-state geography, building standards, mitigation or Risk Rating 2.0 mix changes.
- 2024–2025 are held out of calibration (2024 is immature); open events are not developed.
- Events are FEMA floodEvent labels, not a treaty hours clause; several span more than a week.
- Attritional losses are assumed independent of catastrophe events; paid claims exclude loss adjustment expense.
- Prices are indicative technical premiums with assumed cost of capital, diversification and expenses; they are not market quotes, and the recommendation depends on the diversification assumption.
- Economic capital is an internal 99.5% one-year proxy for this portfolio’s insurance risk only (no reinsurer default, market or operational risk); it is not a Solvency II SCR.
- The portfolio is stylised; market benchmarks support only order-of-magnitude comparison.

Data: OpenFEMA NFIP Redacted Claims and NFIP Redacted Policies. This product uses the FEMA OpenFEMA API, but is not endorsed by FEMA. The Federal Government or FEMA cannot vouch for the data or analyses derived from these data after the data have been retrieved from the Agency’s website(s). The Gulf Coast portfolio is a stylised insurance portfolio calibrated to the exposure and claims experience observed in the NFIP Gulf states; it is not intended to reproduce the finances of the NFIP itself. Reinsurance prices are indicative technical premiums, not market quotes. Economic capital is a 99.5% one-year proxy, not a Solvency II SCR.